Analyst Dashboard

Logic & Calculation Document

LabWise LIMS

7 April 2026

Stats Strip (Top Bar)

Assigned Today

How many tests were assigned to this analyst today.

Logic: Count BookingTest records where assignedTo = this analyst AND createdAt = today.

Completed

How many tests this analyst finished today.

Logic: Count BookingTest records where assignedTo = this analyst, status is completed/reviewed/approved, AND updatedAt = today.

Pending

Tests waiting in the analyst's queue right now.

Logic: Count BookingTest records where assignedTo = this analyst, status is pending or in_progress. No date filter.

Overdue

Tests that have crossed their deadline and are still not done.

Logic: Count BookingTest records where assignedTo = this analyst, status is pending/in_progress, AND dueDate < current time.

On-Time Rate

Percentage of tests finished before their deadline this month.

Logic: (tests completed before dueDate this month / total completed this month) x 100.

All stats refresh every 60 seconds automatically.

KPIs (5 Monthly Metrics)

All KPIs are auto-computed from database records. No manual input needed.

KPI	Target	What It Measures
Samples Completed / Month	120	Volume of work done this month
Average TAT	24 hrs	Time from test reaching dept to result entry
On-Time Delivery	95%	Percentage of results entered before deadline
Rejection Rate	2%	How often reviewer sends results back
Pending Tasks	0	Tests still in the queue

Detailed Calculation Logic

1. Samples Completed / Month

Count all BookingTest records assigned to this analyst with status completed/reviewed/approved, updated this month.

Target: 120 tests (currently same for all, will be per-department later).

Color: Green >= 100, Yellow >= 80, Red < 80.

2. Average TAT (Turnaround Time)

Measures how long the analyst takes from receiving a test to entering the result.

- Start: BookingTest.createdAt — when the booking was done and test landed in the department
- End: Result.enteredAt — when the analyst submitted the result
- Formula: Average of (Result.enteredAt - BookingTest.createdAt) in hours for all completed tests this month
- Fallback: If no Result record exists, uses BookingTest.updatedAt

Color: Green <= 24h, Yellow <= 36h, Red > 36h. Lower is better.

3. On-Time Delivery

Percentage of results entered before the deadline.

- For each completed test: check if Result.enteredAt <= BookingTest.dueDate
- Count on-time / total completed x 100
- Same fallback as TAT if no Result record

Color: Green >= 90%, Yellow >= 70%, Red < 70%.

4. Rejection Rate

How often the reviewer rejects and sends results back.

- Formula: rejected / (completed + rejected) x 100
- Lower is better

Color: Green <= 3%, Yellow <= 7%, Red > 7%.

Current limitation: Only counts tests currently in "rejected" status. Once ResultHistory table is implemented, every rejection will count permanently.

5. Pending Tasks

Count of tests with status pending or in_progress assigned to this analyst.

Target: 0. Color: Green <= 5, Yellow <= 10, Red > 10. Lower is better.

How KPI Progress Bar is Calculated

- Normal KPI: min(100, actual / target x 100)
- Inverse KPI (target = 0): actual is 0 = 100%, otherwise max(0, 100 - actual x 10)
- Inverse KPI (target > 0): min(100, target / actual x 100)

KRAs (4 Weighted Performance Areas)

The overall KRA score is a weighted average. Each area gets a score out of 100, multiplied by its weight.

KRA	Target	Weight	How Score is Calculated
Testing Accuracy	98%	30%	(total tests - rejected) / total tests x 100
TAT Compliance	95%	25%	Same as On-Time Delivery KPI percentage
Sample Throughput	120/mo	25%	min(100, tests completed / 120 x 100)
Quality Score	95%	20%	100 - (rejection rate x 5)

Overall KRA Score Formula

(Accuracy Score x 0.30) + (TAT Score x 0.25) + (Throughput Score x 0.25) + (Quality Score x 0.20)

Example:

- Testing Accuracy: 98.2% !' Score: 100% (exceeds 98% target)
- TAT Compliance: 93.2% !' Score: 98.1% (93.2/95 x 100)
- Sample Throughput: 98/120 = 81.7% !' Score: 81.7%
- Quality Score: rejection 1.8%, so 100 - 9 = 91% !' Score: 95.8% (91/95 x 100)

Overall = (100 x 0.30) + (98.1 x 0.25) + (81.7 x 0.25) + (95.8 x 0.20) = 30 + 24.5 + 20.4 + 19.2 = 94.1%

Color Coding:

- Green: Score >= 90%
- Yellow: Score >= 75%
- Red: Score < 75%

Quality Score Penalty Explained

Each 1% of rejection rate removes 5 points from the quality score:

Rejection Rate	Quality Score	Impact
0%	100	Perfect quality
1%	95	At target
2%	90	Slightly below target
5%	75	Needs attention (amber zone)
10%	50	Critical (red zone)

The x5 multiplier is pending final confirmation from stakeholder.

Work Queue Logic

Sorting

Tests are sorted in two levels:

- First by priority: Critical > Urgent > Normal
- Within same priority: least time remaining comes first

Auto-Escalation

The system automatically bumps priority when TAT is running low:

Condition	Action	Example (48h TAT test)
Less than 25% TAT remaining, currently Normal	Escalate to Urgent	12 hours left !' Urgent
Less than 10% TAT remaining, not already Critical	Escalate to Critical	4.8 hours left !' Critical
Past due date	Show as OVERDUE in red	Deadline passed !' red label

Time Remaining Colors

Time Left	Color	Meaning
More than 4 hours	Green	Comfortable
1 to 4 hours	Yellow	Getting tight
Less than 1 hour	Red	Urgent attention needed
Past deadline	Red + OVERDUE label	Missed deadline

Due Date Calculation

Current: Booking date + TAT hours from test master (e.g., 48h TAT = due in 2 days).

Planned: Smart scheduling that considers lab queue, machine availability, and chemical stock.

Alerts

OOS (Out of Specification)

- Triggers when: Any result parameter entered by this analyst has passFail = "fail"
- Only shows if result status is "entered" or "reviewed" (not fully resolved yet)
- Shows: Sample code, test name, parameter, obtained value, spec limit
- Refresh: Every 30 seconds (most frequent on the dashboard)
- Display: Red banner at top of dashboard

Low Stock

- Triggers when: Any consumable in the analyst's location has current stock <= minimum stock
- Critical flag: Red if stock is zero
- Refresh: Every 5 minutes

Equipment Issues

- Triggers when: Any instrument in analyst's department is not "active"
- Auto-detects: Calibration overdue if next calibration date has passed
- PM due soon: Flagged if calibration due within 7 days
- Only shows instruments with problems; hidden if everything is fine
- Refresh: Every 5 minutes

Auto-Refresh Schedule

The dashboard updates itself automatically. No manual refresh needed.

Data	Refresh Interval	Reason
OOS Alerts	Every 30 seconds	Safety critical — analyst must respond quickly
Stats, Work Queue, Recent Results	Every 60 seconds	Work changes frequently during the day
KPIs, KRAs	Every 5 minutes	Monthly numbers don't change rapidly
Equipment, Consumables	Every 5 minutes	Status changes are infrequent

If the server is unavailable, the dashboard shows the last known data or sample mock data so the page doesn't break.

Data Sources

Where each number on the dashboard comes from:

What	Database Table
Test assignments, status, due dates	BookingTest
Sample codes, client names	Sample, Client (via Booking)
Test names, TAT hours	TestMaster
Department info	Department, DepartmentUser
When the analyst entered the result	Result (enteredAt field)
Pass/fail for OOS detection	ResultParameter
Instrument status, calibration dates	Instrument
Stock levels for consumables	InventoryItem

Pending Items

Item	Current State	What Will Change	Effort
Rejection History	Only tracks current status; rejections lost on re-entry	ResultHistory table logs every rejection permanently	3 hrs
Throughput Target	Hardcoded 120/month for all	Configurable per department by dept head	1 hr
Smart Due Dates	Booking date + TAT hours	Factors in lab queue, machine, chemicals	TBD
Quality Penalty	x5 multiplier set	Awaiting stakeholder confirmation	-